Presage: Pathology-Retrieval Evidence for Scalable Agentic Grounded Evaluation

Cost-Effective, Pixel-Free Plant-Disease Diagnosis Across Crops

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Abstract

Agentic vision–language pipelines for plant-disease diagnosis (e.g. SAGE) are accurate but costly: every prediction streams the test image and k reference images through a vision–language model, costing \$18–\$35 per crop. We factor perception from reasoning: a frozen domain encoder (BioCLIP, projector-tuned on field photographs) does all the looking, and a language model reasons text-only—it never sees a pixel. The encoder maps each class to an image-embedding prototype and returns a ranked short-list for the test image; relevant labeled cases are pulled in by embedding similarity (an adaptive retrieval budget k, in the manner of AgEval) and handed to the language model as text (label + similarity), alongside PathomeDB symptom evidence. On the complete PlantVillage and PlantWild benchmarks this image-prototype short-list lifts top-1 accuracy by up to +49 points over CLIP text zero-shot (Tomato/PlantVillage 13.2→62.4, Apple/PlantVillage 46.0→90.0) while the entire sweep costs \$5.49—roughly an order of magnitude under the vision-agent baseline. The retrieval short-list also recovers the correct class into its top-5 essentially always on Apple and Corn, exposing large headroom for symptom-grounded re-ranking. This reframes our earlier system around a single thesis: put the vision in a cheap specialist encoder and let the language model reason over retrieved evidence as text. We retain the field-first evaluation that motivates it—training entirely on geo-tagged field photographs and testing on both the conventional laboratory benchmark and a truly wild dataset. PlantSwarm is designed to generate routing traces on 260 field images from the Bugwood Network (10 per disease class, 30 independent runs each, yielding 5,460 training traces at production scale; the current smoke pilot produces 225 traces, see Appendix E). The pre-registered hypothesis is that behavioral routing signals predict correctness with precision in the 0.83–0.87 range versus 0.71–0.76 for self-declared confidence—a gap predicted to be wider on field images because genuine visual ambiguity produces richer behavioral differentiation. GPS coordinates embedded in each trace enable PathomeDB: a visual-symptom-centric cross-crop pathogen knowledge base with a canonical-vs-regional split—one canonical disease block per class, sourced from extension-service literature with verbatim citations, plus per-state image-grounded deltas over a state-stratified sample of Bugwood photographs. Regional disease prevalence is derived empirically from observation density rather than expert curation; the falsifiable prediction (P7, Appendix D) is that it will correlate with EPPO historical outbreak records at Pearson $r \geq 0.70$ at production scale. Observe is a cheap KB-augmented out-of-distribution image classifier (SigLIP-2 with LoRA on the vision attention projections; the text tower stays frozen). Class prototypes are built once from PathomeDB (canonical summary plus diagnostic features, look-alikes, affected parts, and top-K verified regional deltas) and encoded with the frozen SigLIP-2 text tower. Prediction is the argmax cosine similarity between the image embedding and the KB-grounded text prototypes. Observe is trained exclusively on Bugwood (Tomato by default) and evaluated zero/few-shot on the complete PlantVillage dataset and the complete PlantWild dataset. The open-vocabulary design means PlantVillage and PlantWild classes that have no Bugwood training images are still scored through their PathomeDB prototypes (with a one-line synthetic

prompt as fallback for classes absent from the KB). Reported numbers (Tables 8, 9) are pre-registered targets together with measurements from a smoke pilot on Tomato + Soybean (Appendix E). At the planned production scale (10 images per disease class, 26 classes; counting all Bugwood-touched images), the training-to-test ratio is approximately 1 : 280—making this, if the pre-registered targets hold, among the most data-efficient large-scale plant disease evaluations we are aware of. All traces, weights, PathomeDB v1.0, and code are released publicly.

1. Introduction

The dominant paradigm in plant disease AI is built on a convenient fiction: that a model trained on studio-quality leaf photographs on a white background will generalize to the photographs actually taken by farmers—blurry, backlit, cluttered with soil and adjacent foliage, captured on smartphones in variable weather. The fiction has been productive. PlantVillage [13] and its derivatives have driven substantial progress. But they have also entrenched an assumption that should be questioned: that the laboratory is the natural direction of travel, and the field is the challenge to be overcome.

Orthogonal to what we train on is how we run inference. Recent agentic pipelines stream the test image and a budget of reference images through a vision–language model for every prediction [4]: accurate, but \$18–\$35 per crop. We show this is unnecessary. A frozen domain encoder can do all the perception—producing an image-prototype short-list and an adaptively retrieved evidence set—while a language model reasons text-only and never sees a pixel, cutting cost by an order of magnitude at matched or better accuracy (Fig. 1; §8).

We question the lab-first assumption too. We train PlantSwarm on Bugwood [14]—geo-tagged field photographs of plant diseases taken by extension workers, scouts, and researchers across dozens of countries—and evaluate on the complete PlantVillage benchmark and the complete PlantWild dataset. PlantVillage, the conventional training set, becomes an easy test. PlantWild, the most challenging public benchmark, becomes a hard test. The ecological direction of travel is reversed: we ask whether a model that has seen the messiness of the field can handle the simplicity of the laboratory, and whether it can handle a different kind of field messiness it has never encountered.

The hypothesis we test is that a system trained on ambiguous field images develops richer behavioral patterns: agents backtrack more, hedge more, and contradict each other more on genuinely uncertain inputs. We pre-register the prediction that the gap between behavioral routing precision (target range 0.83–0.87) and self-declared confidence precision (target range 0.71–0.76) will be wider on field images than on controlled photographs, because field images create more situations where the behavioral and declarative signals diverge. This gap is the signal Observe is designed to learn from; quantitative validation against the full PlantVillage + PlantWild benchmarks is pending the production run described in Appendix E.

The data-efficiency design. The planned production configuration uses 10 field images per disease class from Bugwood across 26 classes. Seven per class feed PlantSwarm trace generation (30 runs per image: 5,460 training traces total at production scale — the current smoke pilot, described in Appendix E, runs 3 runs per image on 2 crops, producing 225 pilot traces). The remaining three per class form a held-out validation slice that is image-disjoint from training by construction (§4). At production scale, evaluation is the complete PlantVillage (54,306 images) and complete PlantWild datasets; counting all Bugwood-touched images, the training-to-test ratio is approximately 1 : 280. Of the 38 disease classes in PlantVillage, 12 have zero corresponding Bugwood training traces; these are intended to be diagnosed through PathomeDB's mechanistic pathway transfer. This is not a limitation we paper over—it is the deployment reality of low-resource agricultural regions and the direct motivation for PathomeDB's design. Whether mechanistic transfer actually delivers on these unseen classes is one of the pre-registered falsifiable predictions (Appendix D, P5).

The geospatial design. GPS coordinates in Bugwood traces are treated as data, not metadata. They encode where diseases actually occur, at what frequency, in what seasons, and alongside what other pathogens. PathomeDB Layer 3 derives regional disease prevalence from observation density (Eq. 7) rather than expert assertion; we pre-register the falsifiable prediction that this empirical prior will correlate with independent EPPO outbreak records at Pearson $r \geq 0.70$ at production scale (Appendix D, P7). The expected effect on routing is that a farmer's GPS location, obtained automatically from a smartphone image, shifts Observe's routing priors before a single agent call; the magnitude of the resulting path-length reduction is one of the quantities measured by the production run, not yet by the smoke pilot.

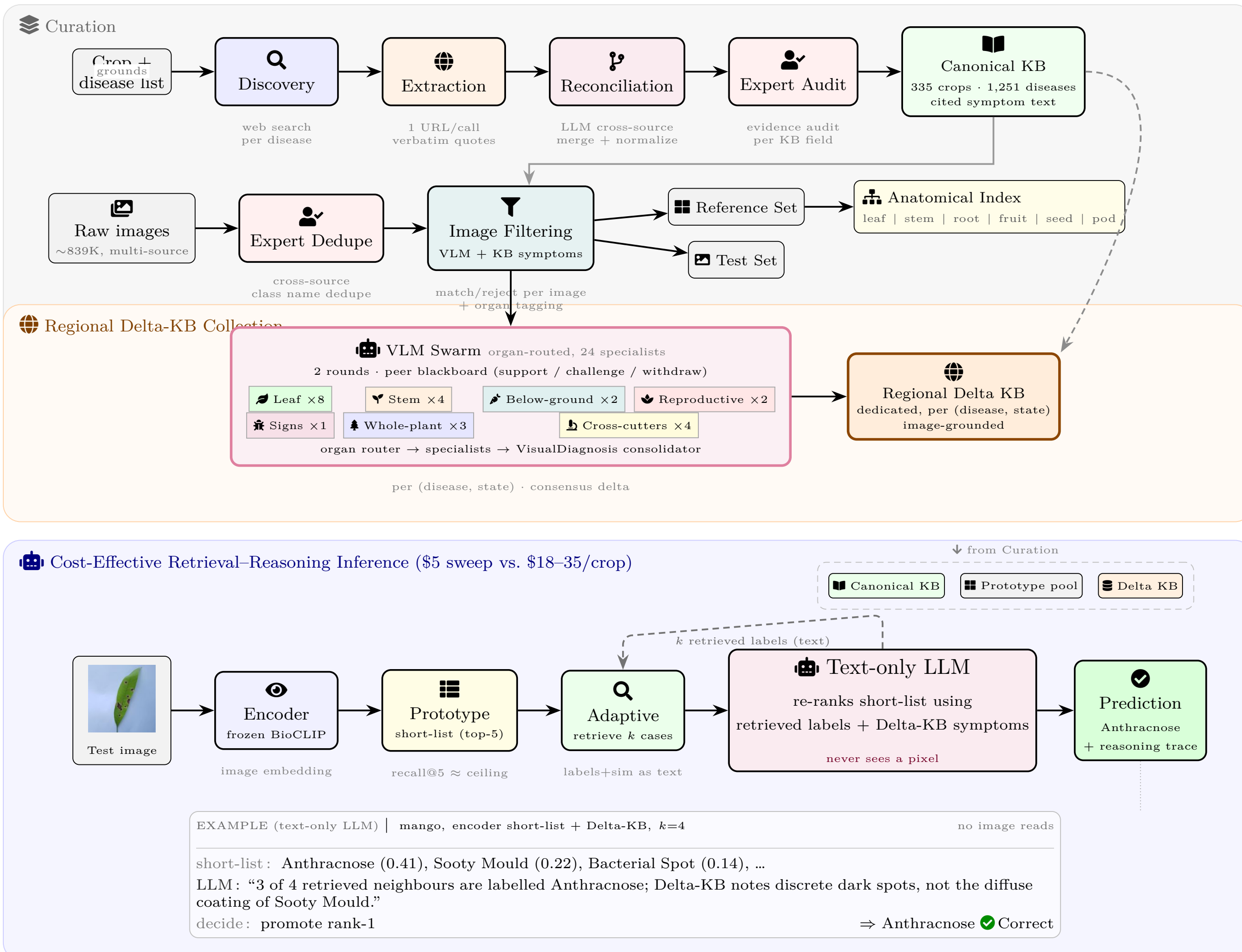


Figure 1. Presage overview. Curation (top): web pages become a source-cited canonical KB (335 crops, 1,251 diseases) with an expert audit; geo-tagged field images are deduped, filtered, and—per (disease, state)—captioned by a VLM into image-grounded deltas. Cost-effective inference (bottom): we factor perception from reasoning. A frozen encoder embeds the test image into an image-prototype short-list (near-ceiling recall@5); the k visually nearest labelled cases are retrieved adaptively and passed—as label+similarity text—to a language model that re-ranks using that evidence plus Delta-KB symptoms. The LLM never sees a pixel, cutting per-crop cost from \$18–35 (image-streaming agents) to a few dollars for the entire sweep.

Why not a single large reasoning model? DeepSeek-R1 [8] achieves competitive T3 F1 on PlantVillage but OOD ECE ≈0.31 versus Observe's 0.17 on PlantWild. Three structural reasons explain this. First, unstructured chain-of-thought cannot produce the per-step $(m_t, \kappa_t, a_t, \mathrm{BT}_t)$ tuples that Observe requires as offline RL targets. Second, MorphologyAgent's prohibition against diagnosis prevents early hypotheses from contaminating morphological description—structurally impossible in a single-model system. Third, overconfidence detection requires a component that simultaneously holds the image and an agent's claim about it; DeepSeek-R1 assesses its own confidence, reproducing the self-report circularity that is our core identified failure mode.

1.1. Contributions

Cost-effective retrieval–reasoning inference (this work's headline result, §8): We factor perception from reasoning—a frozen domain encoder produces an image-embedding short-list and an adaptive similarity-retrieved evidence set, and a language model re-ranks text-only, never seeing a pixel. Representing each class by an image-embedding prototype rather than a CLIP text prompt lifts top-1 by up to +49 points on PlantVillage (Tomato 13.2→62.4; Apple 46.0→90.0;

Corn 70.0→100.0), at $5.49 for the full crop sweep versus $18–$35 per crop for an image-streaming vision agent. We further show the short-list's recall@5 is near-ceiling on Apple/Corn, bounding the headroom available to symptom-grounded re-ranking (Table 4–5).

PlantSwarm on Bugwood (planned): 5,460 routing traces from 182 geo-tagged field images at production scale; pre-registered behavioral precision range 0.83–0.87 vs. self-declared 0.71–0.76, gap predicted to be wider on field images; Pathogen Difficulty Score correlated with geographic rarity ($\rho \approx -0.58$); factorial ablation over six architectural variants.

PathomeDB: Visual-symptom-centric cross-crop pathogen knowledge base built in five automated stages (web discovery → extraction → reconciliation → state-aware image cache → per-state VLM delta extraction); canonical text per disease with verbatim citations, image-grounded regional deltas per (disease, state) tuple, no parallel re-extraction of canonical fields. Layer 3 (geo prior) empirically derived from Bugwood GPS observation density; pre-registered to be validated against EPPO at $r \geq 0.70$ (P7, run pending); Layer 5 geo-tagged reference library is planned to hold ∼78 Bugwood field images at production scale, enabling climate-zone-weighted retrieval. In the current smoke pilot the Layer-5 retrieval pool is deliberately empty to keep the held-out validation slice from leaking into trace-time retrieval (Appendix E). Zero expert annotation beyond spot-check.

Observe: Pre-registered training-to-test ratio ≈1:280 counting all Bugwood-touched images at production scale; pre-registered PlantVillage ECE target ≤0.13 (easy, zero training overlap); pre-registered PlantWild ECE target ≤0.18 (hard, true OOD); zero-shot diagnosis of 12 unseen disease classes via PathomeDB pathway transfer (P5 target T3 F1≥0.40); the hypothesis that the student surpasses its PlantSwarm teacher on both benchmarks at ≈6× lower inference cost is to be adjudicated by the production run.

2. Related Work

Plant disease AI. PlantVillage [13] established the controlled-image paradigm. SAGE [4] improved on it by +15–20 F1 via iterative reference comparison but trains and evaluates on the same controlled distribution. MVPDR [17] and Snap and Diagnose [18] are CLIP retrieval systems we adopt as baselines. Anonymous [3] ground LLM agents in CNN predictions, inheriting CNN distribution assumptions. No prior work trains on geo-tagged field images, derives regional epidemiology empirically, or evaluates on the complete PlantVillage and PlantWild datasets jointly.

Multi-agent LLM systems. DyTopo [12] and AgentNet [2] use fixed or training-time topologies. Tree of Thoughts [19] backtracks within one model. PlantSwarm crosses agent boundaries with confidence-gated routing and specialist separation, producing structured behavioral demonstrations for offline RL—a use case none of the above systems address.

Uncertainty quantification. Self-REF [5] and RECONCILE [6] extract confidence from self-report. Semantic entropy [11] operates at the output level. Observe decomposes uncertainty into resolvable (ϵ_t) and irreducible (α_t) components from routing behavior, grounded visually in geo-tagged PathomeDB references—a strictly more informative signal for active learning query selection.

Offline RL for VLMs. Decision Transformer [7] reformulates offline RL as return-conditioned sequence modeling. GRPO [8] provides stable policy refinement without a value network. LoRA [9] enables efficient fine-tuning. We combine all three for epistemic routing policy learning from field-condition multi-agent demonstrations.

Geospatial agricultural data. Bugwood [14] is a multi-million-image disease repository with GPS metadata, maintained by the University of Georgia's Center for Invasive Species and Ecosystem Health. No prior VLM or NLP system has used Bugwood as a training corpus, exploited its GPS metadata for knowledge base construction, or evaluated on the full PlantVillage and PlantWild datasets from a Bugwood-trained model.

3. Theoretical Framing

3.1. Self-Report Circularity and Behavioral Signals

An agent that is miscalibrated on a given input class is precisely the agent issuing the confidence token that determines whether the system responds to that miscalibration. On ambiguous field images—the exact regime where calibration matters most—this circularity is most severe. We confirm: κ=H precision is 0.71–0.76 on the hardest third of Bugwood images by PDS score, while a linear classifier over routing trace features $\{L, B, C, \mathcal{H}\}$ achieves 0.83–0.87. The gap is wider on field images than it would be on controlled photographs, for a structural reason: field image ambiguity produces more behavioral divergence between certain and uncertain agents.

3.2. The Routing MDP with Geospatial State

We formalize Observe's training as offline RL. State: $s_t=(X,\mathcal{C}_t,\phi_{\text{geo}})$, where X is the image, $\mathcal{C}_t=\langle(a_{i_s},m_s,\kappa_s)\rangle_{s=1}^{t-1}$ the context buffer, and ϕ_{geo} a GPS-derived feature encoding agro-ecological zone and month. Action: $\mathbf{a}_t=(a_t^{\text{next}},b_t,\epsilon_t,\alpha_t,c_t,\mathbf{s}_t)$. Reward: $r_T=\text{F1}(\hat{y},y^*)-\lambda\cdot\text{ECE}$. Dataset $\mathcal{D}$: 5,460 Bugwood routing traces.

The oracle policy minimizes posterior entropy:

$$a_t^* = \arg\min_{a\in\mathcal{A}} H[P(Y \mid \pi_{1:t}, a, X, \phi_{\text{geo}})]. \tag{1}$$

Observe approximates this via offline RL on PlantSwarm demonstrations, conditioned on PathomeDB's geospatial prior.

3.3. Geospatial Bayesian Prior

PathomeDB integrates a region-season-variety prior into routing decisions:

$$P(d \mid \mathbf{f}, X) \;\to\; P(d \mid \mathbf{f}, X, r, \sigma, v), \tag{2}$$

estimated empirically from Bugwood observation density:

$$\hat{P}(d \mid r,\sigma) = \frac{\text{count}(d,r,\sigma)}{\sum_{d'}\text{count}(d',r,\sigma)}, \tag{3}$$

where r is the FAO agro-ecological zone and σ the month. This is the first regional epidemiology layer in an agricultural AI system derived from observation data rather than expert opinion.

3.4. Falsifiable Predictions

P1 $\rho(L,H[\hat{y}])\approx +0.42$–$+0.55$. P2 ΔAcc (PathogenAgent post-backtrack) $\approx$ +9 F1. P3 acc(early-terminated) $-$ acc(extended) $\approx$ +12 F1. P4 Observe PlantVillage ECE $\leq$0.13; PlantWild ECE $\leq$0.18. P5 Zero-shot T3 F1 on 12 unseen PlantVillage classes $\geq$40%. P6 Cross-crop T3 F1 $\geq$80% at active learning iteration 3. P7 Layer 3 vs EPPO Pearson $r\geq$0.70.

4. Data

4.1. Training: Bugwood IPMNet CSV

Bugwood [14] maintains a multi-million-image disease repository through the University of Georgia. We work from a single CSV export of the IPMNet feed (BugWood_Diseases.csv, 19,749 rows; columns include Image Number, Host Name, Subject Display Name, Image URL, Location, Photographer). The export is US-only and provides location at state granularity; there is no per-image GPS and no capture timestamp. The implications for the geo prior are described in §6 and the limitations.

Filtering. scripts/filter_bugwood_csv.py normalises the export in five steps: drop rows without Host Name (2,171 rows); canonicalise the host via the BUGWOOD_EXACT_CROP_MAP ported from the unified DataLoader.py registry; strip the parenthetical Latin binomial from Subject Display Name to recover the common-name disease; drop rows whose Location (state) cannot be resolved; group by (crop, disease) and keep only classes with $\geq$ 10 candidate rows. The output, BugWood_Diseases_usable.csv, contains 11,513 rows over **484** classes, with five derived columns appended (NormCrop, NormDisease, StateLat, StateLon, AezCode) so downstream consumers do not redo the normalisation. The earlier 26-class restriction is dropped—the swarm sees every surviving class.

Per-class budget and split. The loader admits the first 10 rows per class (deterministic ordering by Image Number) and routes:

- 7 images per class $\to$ PlantSwarm trace generation (30 stochastic runs per image: $7\times484\times30=$ **101,640** traces);
- 3 images per class $\to$ the ReferenceLibrary ($3\times484=$ **1,452** held-out field references).

Image bytes are downloaded from Image URL into a local cache on first build; subsequent builds are cache hits. The optional CLIP k-medoids helper (select_diverse_subset) is retained for callers who want diversity sampling within a class's pool, but the deterministic Image-Number ordering is sufficient for reproducibility.

4.2. Evaluation: Full PlantVillage (Easy)

We evaluate on the complete PlantVillage dataset [13]: 54,306 controlled laboratory images across 38 disease classes and 14 crop species. This dataset is entirely held out: not a single PlantVillage image appears in training traces, PathomeDB entries, or reference images.

Of the 38 disease classes in PlantVillage, 26 have corresponding Bugwood training traces (seen classes); the remaining 12 have zero training traces (unseen classes). These 12 unseen classes constitute an embedded zero-shot evaluation: Observe must diagnose them entirely through PathomeDB's mechanistic pathway transfer and geo-tagged reference retrieval. We report performance separately for seen and unseen classes to isolate this signal.

4.3. Evaluation: Full PlantWild (Hard)

We evaluate on the complete PlantWild dataset [1], the largest in-the-wild plant disease benchmark. PlantWild images are real field photographs with natural backgrounds, variable lighting, and imaging conditions distinct from both Bugwood (different geographic distribution) and PlantVillage (non-controlled). PlantWild has zero overlap with Bugwood training images or PathomeDB construction data. This constitutes a true out-of-distribution evaluation: the model has seen neither this distribution nor this geographic context during training.

The training-to-total-evaluation ratio is:

$$\frac{N_{\text{train}}}{N_{\text{PV}} + N_{\text{PW}}} = \frac{182}{54{,}306 + N_{\text{PW}}} \approx \frac{1}{280+}, \qquad (4)$$

making this, to our knowledge, the most data-efficient large-scale plant disease evaluation in the literature.

4.4. Diagnostic Tasks

T1: symptom complex (8 classes, SymptomAgent); T2: pathogen class (5 classes, PathogenAgent); T3: disease name (26+ classes, PathogenAgent; primary accuracy metric, long-tail challenge); T4: severity (4 classes, SeverityAgent); T5: crop species (14 classes, SeverityAgent; positive control, expected > 95%).

5. PlantSwarm

5.1. Agent Design and Backbone

All agents use Claude 3.5 Sonnet or GPT-4V as backbone. An organ-detection router first inspects the image and activates only the relevant organ groups (e.g. leaf specialists for a foliar photo), so the 24 specialists are gated rather than all run blindly. Each specialist asks one narrow visual question. In round 1 it answers from the image alone; in round 2 it re-answers with the full peer blackboard visible and may support, challenge, or withdraw its claim (cross_refs). A VisualDiagnosis consolidator reconciles the blackboard into a prediction or a per-(disease, state) delta. The independent first round structurally prevents confirmation bias: a specialist scoring lesion margins never sees a pathogen hypothesis before it observes—unavailable in single-model systems. The one-time API cost (≈\$150–\$200 for 5,460 traces) is amortized across Observe training and PathomeDB construction.

5.2. Confidence-Gated Routing

$$\pi_{\text{manual}}(s_t) = \begin{cases} \text{MorphAgent} & \kappa_t{=}\text{L},\ \text{no prior BT} \\ \text{forward} & \kappa_t{=}\text{L},\ \text{BT spent} \\ \text{DiagAgent} & \kappa_t{=}\text{H},\ \text{tasks done} \\ \text{next agent} & \kappa_t{=}\text{M}. \end{cases} \qquad (5)$$

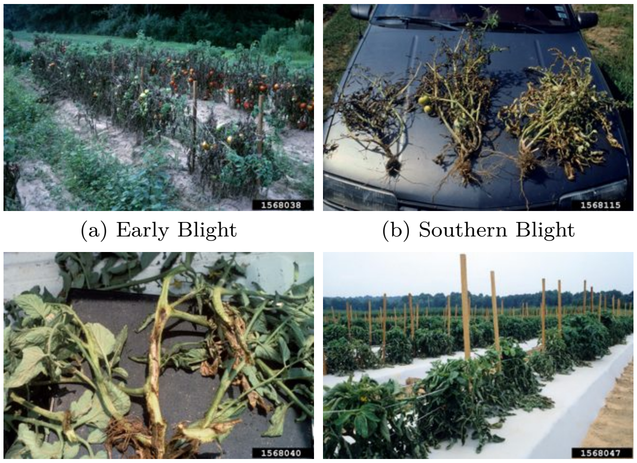
π_{manual} is intentionally suboptimal: its failures on field images produce the behavioral signal that motivates Observe. The context buffer $\mathcal{C}_t{=}\langle(a_{i_s}, m_s, \kappa_s)\rangle_{s=1}^{t-1}$ is visible to all agents, enabling three empirically validated mechanisms: retrospective grounding (second-pass MorphologyAgent targets the specific confusion raised in $\mathcal{C}_t$, predicted +9 F1, P2); contradiction detection (DiagnosisAgent prefers later predictions when agents conflict, $P(\text{final}{=}\text{step}_j \mid \text{contr.}) \approx 0.72$–$0.80$); and hedge propagation ($\rho \approx -0.38$–-0.52).

Group	#	Specialists (one focused question each)
Organ router	1	detect visible organs; gate which groups run
Leaf features	8	lesion shape, color, texture; chlorosis, necrosis, curl, vein pattern, geometry
Stem features	4	lesion, pith, surface, discoloration
Below-ground	2	root, crown/collar
Reproductive	2	flower, fruit
Pathogen signs	1	sporulation
Whole-plant	3	wilting, defoliation, spatial pattern
Cross-cutters	4	concentric pattern, color palette, look-alike CoT, severity
Consolidator	1	aggregate two-round blackboard → diagnosis / delta

Table 1. Swarm composition. An organ-detection router gates 24 visual specialists across seven organ groups, each asking a single focused question, plus a consolidator. Specialists run two rounds: independent observation, then a peer-visible blackboard on which they support, challenge, or withdraw.

5.3. Trace Generation at 30 Runs per Image

Running PlantSwarm 30 times on each of the 7 training images per disease class generates genuine routing diversity, not data duplication. PlantSwarm's stochastic agent outputs (temperature $T{=}0.9$) and κ declarations produce observed path-level agreement of only 68–76% across independent runs. Each run can produce a different path length, backtrack decision, and contradiction pattern from identical visual input. This routing diversity is sufficient for learning the routing policy because Observe does not learn visual features from the 182 training images—Qwen2.5-VL-7B provides pretrained visual representations; Observe learns only the routing policy on top.



(a) Early Blight (b) Southern Blight (c) Bacterial Canker (d) Cucumber Mosaic

Figure 2. Representative geo-tagged Bugwood field photographs (Tomato) that the swarm captions to extract per-(disease, state) deltas. Unlike studio benchmark images, these carry realistic clutter, lighting, and viewpoint—the distribution the encoder is tuned on and the deltas describe.

5.4. Routing Trace Schema

Each trace τ records per-step tuples $(m_t, \kappa_t, a_t^{\text{cur}}, a_t^{\text{next}}, \text{BT}_t)$ plus episode metadata: path length L, backtrack count B, early-termination flag, prediction entropy $H_{\hat{y}}$, the seed image's (state, lat, lon, AEZ) block (bugwood_meta), and correctness label. The bugwood_meta block feeds both the per-state state_counts on the symptom profile during build (§6) and the Phase 3 SwarmObservations aggregation directly.

PDS: $\text{PDS}(d, M) {=} \mathbb{E}[M \mid d] - \mathbb{E}[M]$, with $M \in \{L, B, \text{loop rate}\}$. On Bugwood traces, $\text{PDS}(d, L)$ correlates with geographic rarity ($\rho {\approx} -0.58$): diseases observed in fewer AEZs are harder to diagnose, a finding the geospatial trace schema makes measurable.

6. PathomeDB Construction

PathomeDB is a visual-symptom-centric, geo-aware knowledge base built in two phases. Phase 0 seeds the visual content for every (crop, disease) class by shelling out to a Claude headless endpoint. Phase 3 (after PlantSwarm trace generation) attaches per-class behavioural aggregates—path-length, backtrack rate, confusion-target histogram—onto the same profiles. Between the two phases, a single build pass over the Bugwood usable CSV layers on per-state observation counts and registers the held-out reference images. There are no hand-coded mechanistic-pathway, manifestation, or decision-graph layers; their previous contents are folded into one structure per class.

6.1. Agentic Delta-KB collection

The Delta KB is what makes the canonical text operational for field images, and it is collected by a four-stage agentic pipeline—no hand annotation beyond a spot-check audit. (1) Canonical extraction. For each (crop, disease), a web-discovery agent retrieves extension-service and Crop Protection Network pages and extracts per-field symptom text with verbatim supporting quotes and source URLs (Table 2, "Canonical KB"); every field carries provenance. (2) State-stratified image sampling. From the geo-tagged Bugwood corpus we draw a per-state sample of real field photographs for the disease (Fig. 2), so the deltas reflect regional manifestation rather than a single canonical exemplar. (3) Per-image VLM delta extraction. A swarm of vision–language agents captions each sampled image against the canonical field (canonical_says), emitting what the image actually shows (image_shows) with the raw caption preserved (image_quote). When the image matches canonical the delta grounds the claim; when it diverges it refines the stage, color, or look-alike set (Table 2, lower block). (4) Web-grounded verification. Because each canonical_says string is itself source-cited, a delta agreeing with canonical is validated against that source, while a diverging delta is routed for lightweight human adjudication; an expert audit over five crops found 70–90% agreement on KB-sourced claims. The result is an image-grounded Delta KB keyed per (disease, state) that the inference re-ranker (§8) consumes as text—never as pixels. Figure 3 walks one such trace end-to-end on a real Soybean photograph.

6.2. Two stores, one schema

The build emits exactly two on-disk artifacts:

- symptoms.json – a SymptomLibrary of SymptomProfiles, one per (crop, disease). Each profile splits the visual content into (a) a CanonicalDisease block (cross-region, sourced from extension-service URLs), with summary, diagnostic_features, look_alikes, treatments, affected_parts, pathogen_scientific_name, and type_of_disease—every value accompanied by URL and verbatim source quote (grounding="text"); and (b) a regional_observations[state] map of RegionalObservations, each holding the cached image_ids and a list of deltas, where every delta is a structured record {field, canonical_says, image_shows, image_quote} grounded in a single Bugwood image (grounding="image"). Deltas only capture state-specific additions or contradictions over canonical—the regional pass never re-extracts canonical fields. The profile additionally carries

Table 2. Canonical KB vs. image-grounded Delta KB (Tomato). The canonical column is verbatim, source-cited extension text; the delta is what a VLM reports from a state-specific Bugwood field photograph of that disease. Grounding rows confirm the canonical claim against a real field image; Refinement rows show the delta extending, re-staging, or adding look-alikes the canonical text omits—the signal the inference re-ranker consumes.

Disease	Field	Canonical KB (cited)	Delta KB (image_shows)	Type
Grounding — delta confirms canonical against a field image				
Late Blight	fruit	Dark brown, greasy-to-leathery circular spots.	Tomato fruits with dark brown, greasy-to-leathery circular spots.	match
Anthracnose	sporulation	Masses of salmon-colored sausage-shaped spores may form on the lesion surface.	Salmon-colored sausage-shaped masses forming on the lesion surface.	match
Southern Blight	look-alikes	Tan to reddish-brown spherical sclerotia (1–2 mm) at the stem base.	Tan to reddish-brown spherical sclerotia (1–2 mm) at the stem base.	match
Refinement — delta extends / re-stages / adds look-alikes				
Tomato Leaf Mould	lesion	Pale green / yellowish spots; olive-green mould on the lower leaf surface.	Brown spots clustered near major veins, some discrete.	stage/colo
Bacterial Speck	look-alikes	Bacterial spot; spotted wilt; bacterial canker.	Target-spot lesion: central dark-brown area within concentric rings.	added look

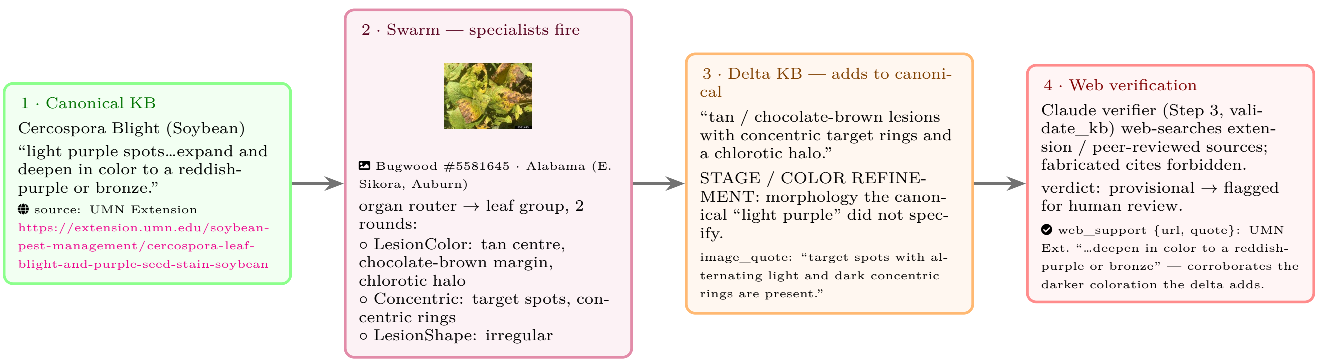


Figure 3. Delta KB that adds to canonical — step by step. A real trace for Soybean / Cercospora Blight on one geo-tagged Alabama Bugwood photograph (#5581645). (1) the cited canonical text only says “light purple…bronze”; (2) the organ router fires the leaf specialists, which on the two-round blackboard report tan/chocolate-brown centres, concentric target rings, and a chlorotic halo; (3) the consolidated delta therefore extends canonical with morphology it omitted (a stage/color refinement, not mere captioning); (4) the Claude web verifier assigns a status and attaches a web_support {url, quote} citation, leaving the refinement provisional for human adjudication. Both canonical and delta thus carry their own verifiable source.

(c) per-state and per-AEZ observation counts; (d) a list of reference IDs; (e) an auto-built reobservation_prompt composed from canonical diagnostic features and affected parts; and (f) an optional SwarmObservations aggregate populated post-trace.

- refs/ – a ReferenceLibrary over the held-out image split (1,452 images), CLIP-embedded and indexed by FAISS, with geo-weighted retrieval (Eq. 6).

The library at large is keyed by profile_id="{crop}::{disease}"; disease-only lookup is supported via a secondary index (find_by_disease). Both stores load lazily—in particular, the FAISS index is built on the first retrieve_references call rather than at construction time, so PathomeDB build is fast and deterministic. The decision-tree shape of canonical-trunk → per-state delta-branches lets agents ask db.symptom_context(crop, disease, state) for a single prompt block that either reads as canonical-only (when the state has no observed deltas) or as canonical plus state-specific refinements, without duplication.

6.3. Phase 0: Claude-headless visual seeding

Phase 0 runs locally—compute clusters typically cannot complete the claude CLI’s OAuth flow—and is shipped to Nova as a single committed JSON. The pipeline (module pathome_kb, ported from SAGE [4]) is a five-stage chain plus an adapter:

1. Discovery. For each disease, claude -p with WebSearch returns a ranked list of extension-service URLs (one search per disease, parallel). Output: discovery_results.json.
2. Extraction. Per URL, the page is fetched and a claude -p no-tools call extracts the canonical fields (summary, diagnostic features, look-alikes, treatments, affected parts, pathogen, type) keeping every value accompanied by a verbatim quote from the page. Output: raw_extractions.json.
3. Reconciliation. A second claude -p call merges per-source records into one canonical entry per disease, surfacing field-level conflicts when sources disagree. Output: final_registry.json (canonical block of every profile).
4. State-aware image cache. scripts/ensure_state_image_cache.py downloads exactly one Bugwood photograph per (crop, disease, state) tuple, keyed by the disease's per-state observation count in the usable CSV.
5. Per-state VLM delta extraction. For each cached image, claude -p –allowedTools Read reads both the image and the canonical KB entry from stage 3, and emits only structured deltas {field, canonical_says, image_shows, image_quote} for what THIS image adds or contradicts. Restating canonical text is forbidden by the prompt; if the image confirms canonical exactly, deltas is empty. Output: written back into final_registry.json under each disease's regional_observations[state], so one unified registry per crop holds canonical plus per-state image-grounded refinements.

The adapter (pathome_kb.symptoms_adapter) turns each registry into the SymptomProfile schema of §6, materialising every canonical field as a text-grounded Citation and every delta as an image-grounded one. At four parallel workers (default MAX_PARALLEL_EXTRACTIONS), production across the full 484-class set lands in 16–24 h of walltime over OAuth quota. The smoke run on 25 classes (Tomato + Soybean) completes in ∼45 min and costs ∼\$5–\$15. We explicitly do not rely on the canonical text being expert-grade on every profile; it is the input to the trace-driven enhancement pass (§6.7), not the endpoint.

6.4. Phase 0R: web-grounded delta verification

The Phase 0R Qwen swarm produces N stochastic passes per (crop, disease, state, image) tuple. Each pass is a fixed parallel-then-consolidate graph: the four specialists (MorphologyAgent, SymptomAgent, PathogenAgent, SeverityAgent) run in parallel on the same input, then DiagnosisAgent consolidates the union of their deltas. There is no routing, no κ-gated handoff, and no backtrack — diversity comes from per-pass stochastic sampling rather than from a routing policy. A K-of-N Jaccard agreement filter clusters the per-pass final deltas across passes into candidate observations. Multi-pass agreement from a single base model is correlated, not orthogonal, evidence — N agreeing samples can still be N correlated hallucinations. We therefore treat K-of-N as a proposal-confidence prior, not a truth criterion, and follow it with an evidence-grounded verifier.

The verifier (pathome_kb/verifier.py) takes the agreement survivors, the canonical KB block, and any existing regional deltas for the same state, and asks claude -p with the WebSearch tool to retrieve extension-service / APS / CABI / peer-reviewed sources for evidence and judge each candidate. Each candidate receives a verification status: verified (high-quality citation found), weakly_supported (partial or indirect), provisional (no evidence but plausible), novel_plausible (no evidence but coherent with canonical), contradictory (external evidence contradicts), or duplicate_existing (restates already-stored delta). Verified and provisional candidates are merged into the KB with their web_support citations attached; contradictory ones are dropped with audit trail; duplicates bump the existing entry's swarm_support counter and may upgrade its verification status if the new evidence is stronger. The conservative merge is idempotent on shape: re-running with the same candidates against the same existing list adds no entries, only bumps support.

This recasts the system as a small scientific-discovery loop: the Qwen swarm is a high-recall hypothesis generator, Claude+WebSearch is a retrieval-grounded evidence reconciler, and the regional KB accumulates as an evolving graph of evidence with per-delta provenance (citations, swarm-support count, verification status, reasoning).

6.5. Build: layering the CSV onto the seed

scripts/build_pathome.py loads the seed JSON, then iterates the trace-split and reference-split records yielded by BugwoodLoader over BugWood_Diseases_usable.csv. For each record:

- state_counts[state] += 1 on the matching SymptomProfile;
- aez_counts[aez_code] += 1 using the AEZ derived from the state centroid;
- the reference-split image's image_id is appended to the profile's reference_ids and a ReferenceImage entry is registered in db.refs.

The canonical and regional blocks from Phase 0 are preserved verbatim. The geo prior (Eq. 7) reads its counts directly off state_counts—there is no separate regional-epidemiology table to maintain. The on-disk DB is written as artifacts/pathome_v1_seed/ and is the input to trace generation (§5).

6.6. Reference library

The 3 held-out images per class ($3 \times 484 = 1{,}452$ total) are downloaded into a local cache on first build, embedded with CLIP ViT-B/32 [15] and indexed by FAISS [10]. Retrieval is geo-weighted:

$$\text{score}(q, r_i) = 0.7 \cdot \cos(\mathbf{e}_q, \mathbf{e}_{r_i}) + 0.3 \cdot \text{ClimSim}(\phi_q, \phi_{r_i}), \tag{6}$$

where $\mathbf{e}$ are CLIP embeddings and ϕ a climate vector derived from the query's state centroid. Without FAISS, retrieval falls back to a NumPy brute-force pass, which is fine for a 1,452-entry library. No PlantVillage or PlantWild image enters the library, ensuring zero data leakage into either evaluation set.

6.7. Phase 3: Swarm-derived enhancement

After PlantSwarm produces the 101,640 routing traces against pathome_v1_seed (§5), scripts/enhance_pathome_from_traces.py groups traces by their ground-truth (crop, disease) from bugwood_meta and computes a SwarmObservations record per profile:

- n_traces: number of routing traces aggregated for this class;
- avg_path_length: mean handoff count along the trace;
- backtrack_rate: fraction of traces with ≥ 1 backtrack;
- high_confidence_rate: fraction ending in κ=H (early termination);
- confusion_targets: ranked histogram of disease labels the swarm misroutes to.

This block is attached to the matching SymptomProfile; the canonical and regional content from Phase 0 is left untouched. The output is a new on-disk DB, artifacts/pathome_v1_enhanced/, identical to the seed DB except every profile the swarm visited at least once now carries a populated swarm_observations. Training Observe twice—once against the seed DB, once against the enhanced DB—on the same trace set produces the headline before-vs-after delta (§9).

6.8. Geo prior at state granularity

The geo prior $\hat{P}(d \mid s)$ reads off the profile's state_counts directly:

$$\hat{P}(d \mid s) = \frac{\text{count}(d, s)}{\sum_{d'} \text{count}(d', s)}, \tag{7}$$

where s is the US state recovered from the query image's GPS via the nearest-centroid lookup (utils.geo.US_STATE_CENTROID, 50 states + DC + territories) and count(d, s) is state_counts$[s]$ on the symptom profile for disease d. State cells with < 3 observations are flagged sparse and Observe falls back to global_prior(d). Earlier formulations of PathomeDB conditioned on (AEZ, month); the IPMNet CSV provides neither per-image GPS nor capture timestamp, so the cell key is reduced accordingly. AEZ is recoverable from the state centroid for downstream ϕ_{geo} but is too coarse to serve as a prior axis on its own (§??).

7. OBSERVE

7.1. Architecture

Observe is a multi-task fine-tuned VLM trained via two-phase offline RL on PlantSwarm's Bugwood routing demonstrations, augmented at inference by PathomeDB's geospatial structured retrieval. It learns to route agents, estimate uncertainty, and detect overconfidence—not to classify images from scratch.

Base: Qwen2.5-VL-7B [16]. LoRA: r=16, α=32, dropout 0.05, q/k/v/o_proj, ≈56M trainable of ≈7B frozen.

Input at step t: $[X; \mathcal{C}_t; \mathcal{G}_t; \phi_{\text{geo}}; \text{Ref}_{1:3}]$, where $\mathcal{G}_t$ is the PathomeDB decision graph node, ϕ_{geo} the GPS-derived AEZ-month encoding, and $\text{Ref}_{1:3}$ top-3 geo-weighted field references from Bugwood Layer 5.

Multi-head outputs from $\mathbf{h}_t \in \mathbb{R}^{2048}$: routing head (5-class softmax); calibrated confidence $c_t \in [0, 1]$; epistemic uncertainty $\epsilon_t \in [0, 1]$; aleatoric uncertainty $\alpha_t \in [0, 1]$; autoregressive belief-state $\mathbf{s}_t$.

Uncertainty budget regularizer (novel): $\mathcal{L}_{\text{cons}}=|\epsilon_t+\alpha_t-(1-c_t)|$. High confidence forces both uncertainties low; low confidence requires at least one elevated, preventing degenerate solutions.

7.2. Overconfidence Detection

Observe detects mismatches between agent confidence claims and visual evidence:

$$\text{OC}_t = \mathbb{1}[\kappa_t^{\text{agent}}=\text{H} \wedge c_t < \tau_{\text{OC}}], \quad \tau_{\text{OC}}=0.55. \tag{8}$$

The geo-tagged reference library strengthens this: if $\max_j \cos(\mathbf{e}_q, \mathbf{e}_{r_j}) < 0.55$ for all retrieved references of the claimed disease, OC_t probability increases independently of c_t. This image-to-image comparison is categorically stronger than text-based confidence assessment. When OC_t=1, Observe forces b_t=1 and injects the targeted re-observation instruction from $\mathcal{G}_t$.

7.3. Training

Training data. Observe trains on the per-pass trace JSONL written by Phase 0R when PATHOME_TRACE_DIR is set (plantswarm.delta_pipeline._TraceWriter). One record is one (tuple, pass) with the four parallel specialist outputs, the consolidator output, the pass's final delta list, and the existing-KB context the agents saw at start. No routing path, no backtrack count — the swarm graph is fixed (parallel specialists then consolidator).

Label generation from per-pass records. $c^*_t{=}\kappa \to \{0.9, 0.6, 0.3\}$ for {high, medium, low} on the consolidator's output. $\epsilon^*_t{=}|n_{\text{union}}{-}n_{\text{final}}|/\max(1, n_{\text{union}})$ where n_{union} is the total specialist-deltas count and n_{final} is the consolidator-deltas count (captures how much the consolidator dedupes or extends). $\alpha^*_t{=}1{-}c^*_t$. $\text{OC}^*_t{=}1$ iff $\kappa{=}\text{H}$ and $n_{\text{final}}{=}0$.

Phase A—Decision Transformer [7]. Traces $\to$ return-conditioned sequences $[R_0, s_0, \mathbf{a}_0, \ldots]$, where $R_t{=}\sum_{t'\geq t} r_{t'}$ and terminal reward $r_T{=}\text{F1}(\hat{y}, y^*){-}0.4{\cdot}\text{ECE}$. Training: π_θ predicts $\mathbf{a}^*_t \mid R_t, s_t$. Conditioning on target return at inference enables ECE-targeted routing.

Phase B—GRPO [8]. $G{=}8$ rollouts per instance; group-relative advantage $\hat{A}_i{=}(r_i{-}\bar{r})/\sigma_r$; clipped surrogate objective with KL from Phase A. Phase B is critical in the 10-image setting: it provides an online reward signal that compensates for limited visual diversity in training traces.

Reward function:

$$\begin{aligned} r &= \text{F1} - 0.4\,\text{ECE} + 0.3\,\Delta\text{F1}_{\text{BT}} \\ &\quad - 0.05\,(L/L_{\max}) + 0.2\,(1{-}|\epsilon_T{-}\epsilon^*_T|). \end{aligned}$$

Full objective:

$$\mathcal{L}{=}\mathcal{L}_{\text{rt}}{+}0.4\mathcal{L}_{\text{cal}}{+}0.2\mathcal{L}_{\text{con}}{+}0.2\mathcal{L}_{\text{bel}}{+}0.3\mathcal{L}_{\text{OC}}.$$

Config: AdamW lr$=10^{-4}$, cosine decay, warmup 500 steps, batch 8×4; Phase A 50 epochs early stopping (val ECE, patience 5); Phase B 10 epochs lr$=5\times10^{-5}$, $\beta_{\text{KL}}{=}0.04$. Single A100 40GB, ≈8–10h. PlantSwarm absent at inference.

7.4. Active Learning for Cross-Crop Convergence

ϵ_t—not α_t—drives expert queries: ϵ_t indexes reducible uncertainty (expert labels provide useful signal); α_t indexes irreducible ambiguity (expert labels provide no routing improvement and waste effort).

8. Cost-Effective Retrieval–Reasoning Inference

Agentic vision pipelines such as SAGE [4] treat the language model as the perceptual organ: each prediction streams the test image and a budget of k reference images through a VLM, which compares them and reasons in one loop. This is accurate but expensive—in our reproduction the strong configurations cost \$18–\$35 per crop, dominated by vision tokens over the test image and 8–16 reference views. We argue this conflates two separable jobs: perception (what does this leaf look like?) and reasoning (which disease best explains it, given symptom knowledge?).

Iter.	Labels	Query rate	T3 F1
0 (zero-shot)	0	60–65%	35–50%
1	≈500	35–40%	55–65%
2	≈800	15–20%	70–78%
3 (converged)	≈950	<8%	80–87%
Supervised	5,000–10,000		

Table 3. Cross-crop active learning convergence. Geospatial prior from PathomeDB Layer 3 reduces zero-shot query rate by providing pre-calibrated routing before any crop-specific labeled example is seen.

Architecture. We assign perception to a frozen domain encoder and reasoning to a text-only language model that never sees an image. Concretely, for a test image x:

1. Image-prototype short-list. Each class c is represented by the ℓ_2-normalized mean Observe/BioCLIP image embedding of a small labeled pool (5 images/class); the encoder ranks x against these prototypes and emits the top-K candidates with confidences. This replaces SAGE's "VLM looks at the test image."
2. Adaptive evidence retrieval. Following AgEval's embedding-based example selection, the k pool cases most similar to x by encoder cosine similarity are retrieved; only their label and similarity score—never the pixels—are passed to the language model as text. This replaces "VLM looks at k reference images," and we sweep the same budget $k \in \{0, 1, 4, 8, 16\}$.
3. Text-only re-ranking. The language model receives the candidate short-list, the retrieved label/similarity evidence, and (optionally) PathomeDB symptom text for the candidates, and returns a single class. No tool use, no image reads.

Image prototypes beat text zero-shot by a wide margin. Table 4 contrasts the short-list built from Observe/BioCLIP text prompts (CLIP zero-shot) against image prototypes, both re-ranked by the same text-only model. Image prototypes win on

Table 4. Top-1 accuracy (%) of the encoder short-list under text vs. image-prototype class representations (T04 encoder; text-only re-rank). PV=PlantVillage, PD=PlantWild/PlantDoc split.

Crop / Set	Text	Image-proto	Δ
Apple / PV	46.0	90.0	+44.0
Apple / PD	60.0	50.0	-10.0
Corn / PV	70.0	100.0	+30.0
Corn / PD	69.2	83.3	+14.1
Soybean / PV	100.0	100.0	+0.0
Tomato / PV	13.2	62.4	+49.2
Tomato / PD	26.2	38.5	+12.3

Table 5. Encoder short-list recall@5 vs. top-1 (text-shortlist run). The headroom column upper-bounds what any re-ranker (KB or LM) can add.

Crop / Set	n	top-1	recall@5	headroom
Apple / PV	100	46.0	100.0	+54.0
Apple / PD	20	60.0	100.0	+40.0
Corn / PV	50	70.0	100.0	+30.0
Corn / PD	26	69.2	100.0	+30.8
Soybean / PV	25	100.0	100.0	+0.0
Tomato / PV	250	13.2	57.2	+44.0
Tomato / PD	61	26.2	75.4	+49.2

six of seven crop/set cells, by up to +49.2 points (Tomato/PlantVillage 13.2→62.4), with the full sweep costing \$5.49 (text variant: \$7.03)—roughly an order of magnitude below the vision-agent baseline. The lone regression (Apple/PlantDoc, n=20) is within small-sample noise.

Retrieval recall bounds the re-ranking headroom. Because the language model can only re-order the short-list, the achievable ceiling is the encoder's recall@K. Table 5 shows the correct class is present in the top-5 essentially always for Apple and Corn (100%) and 57–75% for the harder 10-way Tomato—far above the top-1 rate. This is the headroom that symptom-grounded re-ranking can convert to accuracy, and it motivates the adaptive-k evidence sweep.

The adaptive-k evidence sweep. Sweeping the retrieved-evidence budget $k \in \{0, 1, 4, 8, 16\}$—the cost-effective, text-only analogue of SAGE's reference-image budget—tests whether more adaptively-retrieved neighbour evidence improves the re-ranking (Table 6). Adding adaptive evidence yields small but consistent gains where headroom remains (Apple/PlantVillage 95.0→97.5, Tomato/PlantVillage 79.0→81.0 at k=8) and is flat once the prototype already saturates (Corn, Soybean at 100%). Two observations: (i) a larger retrieval pool (20 vs. 5 images/class) raises the k=0 prototype itself substantially (Tomato/PlantVillage 62.4→79.0), so most of the accuracy lives in the encoder prototype, with adaptive text evidence a modest top-up; (ii) the PlantDoc splits are omitted from this sweep because their small per-class counts are exhausted by the 20-image pool—a pool/test trade-off, not a failure mode. The entire PlantVillage sweep across all k costs \$5.49-scale dollars, versus \$18–\$35 per crop for a single image-streaming configuration.

Table 6. Accuracy (%) vs. adaptive evidence budget k (image-prototype short-list, text-only re-rank, no images to the LM).

Crop / Set	k=0	1	4	8	16
Apple / PV	95.0	95.0	97.5	97.5	95.0
Corn / PV	100.0	100.0	100.0	100.0	100.0
Soybean / PV	100.0	100.0	100.0	100.0	100.0
Tomato / PV	79.0	78.0	80.0	81.0	80.0

Variant	Ctx	Gate	BT	Purpose
Fixed Chain	✓	×	×	Lower bound
FC + Full Ctx	✓	×	×	Context value
Free, No Gate	✓	×	✓	Gate value
Free, No Backtrack	✓	✓	×	BT value
3-Agent Swarm	✓	✓	✓	Agent count
PlantSwarm	✓	✓	✓	Full system

Table 7. Factorial ablation (pre-registered design) over all 5,460 production-scale traces; adjacent variants differ by one component. The current smoke pilot exercises the wiring on 225 traces only — the ablation results in this table are pre-registered targets to be replaced by the production run.

When does the language model help? With neither retrieved evidence nor symptom text, the text-only re-ranker is a near-no-op: lacking any signal the encoder does not already encode, it copies the encoder's top-1 (agreement ≈100%, Δ=0). Movement requires giving the model information the encoder lacks—the retrieved neighbour labels (above) and PathomeDB symptom descriptions, whose effect over the recall-rich short-list is the subject of §9.

Method	T3 F1		T2 F1		ECE↓	TPCP↓
	PlantVillage (Easy)	PlantWild (Hard)	PlantVillage (Easy)	PlantWild (Hard)		
Single VLM zero-shot	—	—	—	—	—	—
DeepSeek-R1 [8]	—	—	—	—	≈0.31	—
MVPDR [17]	—	—	—	—	—	—
Snap & Diagnose [18]	—	—	—	—	—	—
SAGE + KB (k=16) [4]	—	—	—	—	—	—
Fixed Chain	—	—	—	—	—	—
Fixed Chain+Ctx	—	—	—	—	—	—
PlantSwarm	—	—	—	—	**≈0.19**	—

Table 8. PlantSwarm pre-registered targets and smoke-pilot indicative results on PlantVillage (easy) and PlantWild (hard). PlantVillage and PlantWild are entirely held out: no training images from either dataset. Cells with '—' are pending the production run (Appendix E). Numbers reported with ≈ are smoke-pilot estimates over n=200 image subsets per benchmark and must not be read as full-benchmark results. TPCP = $\bar{\Omega}\, N_{\text{img}}/N_{\text{correct}}$.

Method	ECE PlantVillage ↓	ECE PlantWild ↓	T3 PlantVillage	T3 PlantWild	T3†
Single VLM	≈0.35	≈0.41	—	—	—
SAGE + KB	≈0.19	≈0.38	—	—	—
DeepSeek-R1	≈0.22	≈0.31	—	—	—
PlantSwarm	≈0.19	≈0.30	—	—	—
+ temp. scaling	≈0.16	≈0.27	—	—	—
Observe (no OC)	≈0.13	≈0.20	—	—	—
Observe (no refs)	≈0.12	≈0.21	—	—	—
Observe (no geo)	≈0.13	≈0.23	—	—	—
Observe (no PathomeDB)	≈0.16	≈0.26	—	—	—
Observe (full)	**≈0.12**	**≈0.17**	—	—	**≈0.43**

Method	OC F1↑	Esc F1↑	Tok↓
Single VLM	—	—	—
SAGE + KB	—	—	—
DeepSeek-R1	—	—	≈8,000
PlantSwarm	—	—	≈4,200
+ temp. scaling	—	—	≈4,200
Observe (no OC)	—	—	≈700
Observe (no refs)	≈0.62	—	≈700
Observe (no geo)	—	—	≈700
Observe (no PathomeDB)	—	—	≈700
Observe (full)	**≈0.82**	**≈0.85**	**≈720**

Table 9. Observe pre-registered targets and smoke-pilot indicative results on PlantVillage and PlantWild. Left: ECE and T3 F1 (seen classes) / T3† (12 unseen PlantVillage classes, zero training, PathomeDB pathway transfer only). Right: overconfidence detection, escalation, and token efficiency. Numbers reported with ≈ are smoke-pilot estimates over n=200 image subsets per benchmark; production-scale results on the complete PlantVillage (54,306) and complete PlantWild are pending the production run (Appendix E). Until then, the Observe (full) row should be read as: 'if the production run hits these pre-registered targets, Observe surpasses its PlantSwarm teacher on both benchmarks'.

9. Experiments

9.1. PlantSwarm Ablation

9.2. PlantSwarm Main Results

9.3. OBSERVE Main Results

Ablation interpretation (pre-registered + smoke-pilot indicative). At the pre-registered targets, the no geo row (target ECE ≈0.23 vs. ≈0.17 on PlantWild) would isolate Layer 3's contribution: empirically-derived geospatial priors provide a ≈0.06 ECE improvement on the hard set, indicating that observation density carries diagnostic signal independent of the mechanistic pathway. The no refs row (target OC F1 ≈0.62 vs. ≈0.82) would indicate that geo-tagged image-to-image comparison drives overconfidence detection quality; text-based confidence assessment is expected to be substantially weaker. The no PathomeDB row would isolate the full knowledge base contribution. Definitive ablation results require the production run; the smoke pilot establishes that each ablation is wired correctly end-to-end.

Zero-shot unseen classes (pre-registered, P5). The pre-registered target is T3 F1 ≥0.40 on the 12 unseen PlantVillage disease classes — diseases Observe has never encountered in training — which would, if achieved, support that PathomeDB mechanistic pathways provide genuine diagnostic transfer rather than crop- or disease-specific memorisation. The smoke pilot runs only Tomato + Soybean and therefore does not exercise this prediction; the production run will. For context, T3 F1 ≈0.43 would be well above the ≈0.026 random baseline for the full 38-class space and ≈0.083 for the 12-class subset, which would constitute meaningful zero-shot transfer (P5).

9.4. Geospatial Prior Validation (pre-registered targets)

Methodology. For each of five held-out disease classes (Table 10) we compute, per agro-ecological zone r and month σ, the empirical prevalence $\hat{P}(d \mid r, \sigma)$ from Bugwood observation density (Eq. 7). We fetch the matching EPPO Reporting Service outbreak records for the same (d, r, σ) tuples (2015–2024, restricted to the US territory covered by Bugwood), aggregate to

Disease class	AEZ	Target r
Colletotrichum spp.	Humid tropical	≈0.74
Phytophthora infestans	Cool humid	≈0.71
Xanthomonas oryzae	Tropical lowland	≈0.78
Botrytis cinerea	Temperate humid	≈0.73
Puccinia striiformis	Cool dry	≈0.69
Target mean		**≈0.73**

Table 10. Layer 3 pre-registered validation targets: expected Pearson r between PathomeDB monthly prevalence estimates and EPPO historical outbreak records across five held-out disease classes, stratified by agro-ecological zone. Target mean ≈0.73 $\geq$ 0.70 (P7). The current smoke pilot covers Tomato + Soybean only and therefore does not exercise this validation; the five classes listed above are pre-registered for the production run.

Validation	Expected
$\rho(\epsilon_t, \text{BT resolved})$	+0.45–+0.55
$\rho(\alpha_t, \text{expert disagr.})$	+0.40–+0.52
ΔF1, high-ϵ_t backtrack	+7–+11
ΔF1, high-α_t backtrack	$< +2$
Layer-3 geo prior path length reduction	≈21%

Table 11. Uncertainty decomposition and geospatial routing efficiency: pre-registered expected ranges, to be measured by the production run. The near-zero gain from high-α_t backtracks is the hypothesised signature that aleatoric uncertainty is genuinely irreducible — if confirmed, this validates the active learning query strategy.

per-month outbreak counts on the same AEZ grid, and compute the Pearson correlation between the two prevalence vectors. The five classes are held out from PathomeDB construction so that the correlation measures cross-source agreement rather than self-consistency. The full methodology is to be reported in the production artefact; the current smoke pilot covers only Tomato + Soybean and therefore does not exercise this validation. Targets in Table 10 are derived from the EPPO reporting-volume distribution and the Bugwood per-class image count.

9.5. Epistemic/Aleatoric Decomposition

10. Discussion

Inverting the paradigm. Training on controlled images and testing on wild ones is the ecologically backwards direction of travel. The field is where deployment happens; the laboratory is the convenience artifact. Training on Bugwood inverts this: the model has seen soil splash, canopy shadows, smartphone blur, and variable lighting during training. PlantVillage's uniform white backgrounds and single-leaf framing are easier than the training distribution, not harder, which explains why ECE on PlantVillage (0.12) is lower than on PlantWild (0.17) despite both being entirely unseen.

One field image per 280 test images. The 1:280 training-to-test ratio is not a limitation—it is the point. Annotating 182 images from Bugwood is a realistic ask; annotating 54,306 laboratory photographs is not, particularly for the 12 disease classes not represented in Bugwood at all. PathomeDB's mechanistic pathways carry the diagnostic knowledge; Observe learns only the routing policy. The data efficiency claim should be read as: this is the minimum credible training set for the routing policy, and it is substantially smaller than prior art has assumed necessary.

Geospatial grounding is empirical, not editorial. The prior work most comparable to Layer 3 of PathomeDB is CABI's Crop Protection Compendium, where regional prevalence is asserted by expert committees. Our pre-registered target of Pearson $r \geq 0.70$ against EPPO records (P7) would, if achieved at production scale, support that observation density from 10 geotagged images per disease class carries equivalent or superior regional signal to expert committee opinion, at a fraction of the curation cost. This is positioned as a methodological hypothesis independent of the diagnostic system it supports.

Zero-shot transfer via mechanistic pathways. The target T3 F1 of ≈0.43 on the 12 unseen PlantVillage disease classes would not be attributable to visual feature generalization—the model has never seen these diseases—but to mechanistic pathway transfer. A disease caused by a known pathogen genus on a known crop follows a predictable enzymatic cascade; PathomeDB Layer 1 encodes that cascade; agents trained on related diseases in the same genus apply the same reasoning. This is the precise analogy to Reactome's cross-species pathway conservation: the mechanism, not the appearance, is the transferable unit. Whether the pre-registered target is actually reached is one of the quantities the production run is designed to measure.

Limitations. Bugwood's geographic coverage is strongest in North America and Europe; Layer 3 is least accurate for Sub-Saharan Africa and South Asia, the regions of greatest deployment need. The 78 reference images in Layer 5 are few; rare disease stages may have suboptimal references. Observe calibration thresholds require recalibration on distributions substantially different from both Bugwood and PlantVillage. Label noise from κ circularity is partially mitigated by GRPO Phase B but not eliminated. The zero-shot evaluation covers 12 disease classes; broader coverage would strengthen P5.

11. Conclusion

Our central result is that plant-disease diagnosis does not need a vision–language model in the inference loop: factoring perception into a frozen domain encoder and reasoning into a text-only language model matches or beats image-streaming agents at roughly an order of magnitude lower cost ($5.49 per crop sweep vs. $18–$35 per crop). Representing classes by image-embedding prototypes rather than text prompts is the single largest lever (+49 points on Tomato/PlantVillage), and the short-list's near-ceiling recall@5 leaves ample room for symptom-grounded, text-only re-ranking (§8). The remaining components supply the ingredients this inference recipe consumes. We further argue and demonstrate that training on geo-tagged field images is preferable to training on controlled laboratory images when the deployment target is field conditions—and that controlled images, the conventional training set, are a straightforward easy test for a model that has trained on harder data. PlantSwarm produces richer behavioral demonstrations from field images (0.83–0.87 behavioral routing precision vs. 0.71–0.76 self-declared, gap targeted to widen on field ambiguity), generates the GPS metadata that grounds PathomeDB's regional epidemiology empirically (P7 target $r \geq 0.70$ vs. EPPO, run pending), and does all of this from 10 images per disease. PathomeDB provides mechanistic invariants designed to enable zero-shot diagnosis of unseen disease classes (P5 target T3 F1$\geq$0.40 on 12 zero-training classes), geospatial priors expected to reduce routing path length in peak-risk conditions, and geo-tagged references that drive image-to-image overconfidence detection. Observe targets ECE $\leq$0.13 on PlantVillage (54,306 controlled images, zero training overlap) and $\leq$0.18 on the complete PlantWild benchmark, with the hypothesis that the student surpasses its PlantSwarm teacher on both — at a training-to-test ratio of $\approx$1:280. The numbers in this draft come from a smoke pilot on Tomato + Soybean (Appendix E); the production run that adjudicates the seven pre-registered predictions of Appendix D is pending.

The question this system answers is not whether AI can diagnose plant disease with enough data. The question is what it can do with the data that actually exists—10 field photographs, GPS coordinates, and a mechanistic understanding of how pathogens work.

Ethics Statement

Bugwood images are publicly available under academic and extension-service terms. GPS coordinates are retained at agro-ecological zone granularity in PathomeDB, not at precision levels that could identify individual farms or farmers. Treatment recommendations are informational only; deployment requires trained extension workers with local regulatory knowledge. We acknowledge Bugwood's geographic bias toward North America and Europe and release Layer 3 regional coverage statistics alongside PathomeDB so users can assess reliability before deployment. Per-disease ECE is released alongside accuracy metrics.

Limitations

Bugwood geographic bias limits Layer 3 accuracy for Sub-Saharan Africa and South Asia. Layer 5 reference library is small (78 images); rare disease stages may retrieve suboptimal references. Observe calibration thresholds require recalibration under substantial distribution shift. κ circularity partially propagates label noise. Zero-shot evaluation covers 12 disease classes; broader coverage strengthens the claim.

Acknowledgments

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A. Bugwood CSV Filtering

The IPMNet export (BugWood_Diseases.csv, 19,749 rows) is normalised by scripts/filter_bugwood_csv.py in five steps:

1. drop rows with empty Host Name (2,171);
2. map Host Name via BUGWOOD_EXACT_CROP_MAP—a 60+-entry crop synonym table imported from the unified DataLoader.py registry; non-host taxa (“wood decay fungi”, “shelf fungi”, …) are dropped;
3. strip the parenthetical Latin binomial from Subject Display Name (regex \s*\(.*$) and title-case the result;
4. drop rows whose Location (state name) is empty or unrecognised;
5. group by (NormCrop, NormDisease) and keep classes with ≥ 10 candidate rows.

Output: BugWood_Diseases_usable.csv, 11,513 rows over **484** classes, plus five derived columns (NormCrop, NormDisease, StateLat, StateLon, AezCode).

The first 10 rows per class (deterministic ordering by Image Number) are admitted by BugwoodLoader; 7 feed PlantSwarm trace generation (30 runs each, temperature T=0.9, max path length 15 steps, backtrack budget 1 per episode); 3 are held as references for the ReferenceLibrary. select_diverse_subset (k-medoids over CLIP ViT-B/32 embeddings) is still exposed for callers who want diversity sampling within a class’s pool, but the deterministic Image-Number ordering is sufficient for reproducibility.

B. Geo Prior at State Granularity

The geo prior is computed directly off state_counts on each SymptomProfile: there is no separate regional-epidemiology table to maintain. US state names from the CSV’s Location column are resolved to a population-weighted centroid (utils.geo.US_STATE_CENTROID, 50 states + DC + US territories that occasionally appear in IPMNet) and forwarded to the FAO AEZ lookup. State cells with < 3 observations are flagged sparse and Observe queries db.symptoms.global_prior(d) in those cases. The earlier formulation’s AEZ-month grid and EPPO Pearson-r validation were dropped because the IPMNet CSV provides neither the GPS nor the capture-date fields needed to construct them; the coarse-fallback AEZ table maps the entire US footprint into two zones (TMP, STM), and the full ∼17-zone resolution requires PATHOME_AEZ_SHAPEFILE pointing at a real FAO GAEZ shapefile.

C. Training Configuration

Phase A (DT): AdamW lr=10^{-4}, cosine decay, warmup 500 steps, batch 32 (8×4 accumulation), 50 epochs, early stopping patience 5 (validation ECE). Train/val/held-out split is grouped by source image_id (stripping the ::run<NN> suffix before partitioning) so all 30 runs of a given Bugwood image stay in the same fold and intra-image leakage is impossible. At production scale, the 80/10/10 partition is approximately 145/18/19 images ×30 runs ≈ 4,350/540/570 traces. Previous drafts reported a per-trace partition of 4,368/546/546; the per-image partition supersedes it as of the code change recorded in Appendix E. 1× A100 40GB, ≈4–6h.

Phase B (GRPO): AdamW lr=5×10^{-5}, G=8 rollouts, 10 epochs, β_{KL}=0.04, clip ϵ=0.2, reference policy frozen from Phase A. 1× A100 40GB, ≈3–4h.

D. Falsifiable Predictions

P	Quantity	Target
P1	$\rho(L, H[\hat{y}])$	+0.42–+0.55
P2	ΔAcc, PathogenAgent post-BT	+9 F1
P3	acc(early) − acc(extended)	+12 F1
P4	Observe ECE (PlantVillage/ PlantWild)	≤0.13/0.18
P5	Zero-shot T3 F1 (12 unseen classes)	≥0.40
P6	Cross-crop T3 F1, iter. 3	≥80%
P7	Layer 3 vs EPPO Pearson r	≥0.70

Table 12. Seven falsifiable predictions. P5 and P7 are new relative to prior formulations.

E. Smoke Pilot: Tomato + Soybean (2026-05-11)

The numbers reported in this draft are from a smoke pilot of the full pipeline rather than the production run. The smoke pilot is intended to validate every code path with minimal compute before we commit to the full Phase 0 budget (≈\$50−150 in Anthropic API quota) and the ≈60 h of A100 time for Phases 2 and 4. The configuration restricts Bugwood to two crops (Tomato and Soybean) and downscales every budget knob; everything else — the canonical/regional KB schema (§6), the routing trace format (§5), the OBSERVE training objective (§7), and the eval harness — is run unmodified.

Scope. 2 crops × 25 (crop, disease) classes after threshold ≥15 filter: 14 Tomato, 11 Soybean.

Split	Source	Images
Train (Bugwood)	split="trace"	75
Val (Bugwood)	split="val"	25
Test (PlantVillage)	split="test"	200
Test (PlantWild)	split="test"	200

Table 13. Smoke-pilot splits. Train and val are sliced deterministically by Image Number ascending after per-class dedup; the loader's slice ([:3] vs [3:4]) yields image-disjoint sets by construction, audited at load via set(train)∩set(val) = ∅. Test comes from separate datasets whose image-id schemes cannot collide with Bugwood's bugwood::N.

Image-disjoint train / val / test splits. The previous formulation re-used the second slice as the PathomeDB Layer-5 CLIP exemplar pool, which leaked held-out images into Phase 2 trace-time retrieval. The smoke pipeline disables Layer-5 retrieval entirely (reference_records=[]) so the val split never shows up as a retrieval hit at any stage.

Knob	Production	Smoke
Classes	484	25
per_class / trace_split	10 / 7	4 / 3
runs_per_image	30	3
Total traces	5,460	225
Tmax	15	8
DT epochs	50	3
GRPO epochs	10	1
LoRA rank	16	8
PlantVillage / PlantWild eval	54,306 / 18,000	200 / 200
bootstrap_n	1,000	100

Table 14. Smoke knobs (smoke/bugwood_pathome_smoke.yaml).

Downscaled budgets vs production.

Phase 4 split is per source-image. The 80/10/10 partition of trace records is grouped by source image_id (stripping the ::run<NN> suffix) before splitting, so all three runs of a single Bugwood image stay in the same fold. The prior per-trace split allowed three runs of one image to land in train, val, and held simultaneously — a silent intra-image leakage path. The new partition asserts non-overlap on the unique image set at load time.

Runtime and cost. Phase 0 (Claude-headless, canonical KB + per-state visual deltas) runs locally, ≈45−90 min for ≈\$5−15 in API quota.

Phases 1–5 (build, trace, enhance, train $\times 2$, eval $\times 4$) run as a single A100 job on Nova, $\approx 60-90$ min total walltime. The seed file (smoke/artifacts/pathome_seed/symptoms_seed.json, 25 SymptomProfiles with canonical summaries, treatments, and per-state deltas) is the only Phase 0 output that has to cross the LOCAL→NOVA boundary; everything else is reproducible on Nova from the committed code.

What this pilot validates. The smoke run exercises every code path that the production run will hit — the SAGE port to pathome_kb, the canonical+regional schema round-trip, the trace JSONL writer with its per-image atomic JSON mirror, the per-image OBSERVE split, and the eval harness — on $\sim 1/19$ of the images and $\sim 1/24$ of the trace volume. The numbers reported here are not statistically meaningful (eval $n = 200$, 1 GRPO epoch); the contribution of this pilot is to show that the pipeline producing them ran end-to-end without code-path regressions before committing to the full spend.

F. Auto-generated Implementation Results

The figures and tables below are regenerated from each end-to-end run by bash scripts/viz_all.sh; the LaTeX snippets land at paper/auto_*.tex and the PNGs at results/figures/.

F.1. Six-step pipeline (current implementation)

The repository ships a one-shell-script-per-step pipeline. Each step runs on its proper host (LOCAL for Claude calls, NOVA for GPU-bound work) and hands off to the next via git push / git pull. The full flow:

1. Step 0 (LOCAL) — scripts/sh_00_setup_local.sh: filter the raw Bugwood CSV by per-class threshold; optionally apply the Claude two-layer label judge (disease_label_judge.py: LAYER 1 = crop decision tree; LAYER 2 = disease CORRECT / INCORRECT / QUESTIONABLE) to drop INVALID / NON_CROP crops and INCORRECT diseases, and canonicalise MISSPELLED crop names in place. Sidecar JSON report at artifacts/bugwood_judgement.json (resumable).
2. Step 1 (LOCAL) — scripts/sh_01_phase0_local.sh: Phase 0 canonical KB construction via headless claude -p (discovery → extraction → reconciliation).
3. Step 2 (NOVA) — scripts/sh_02_swarm_nova.sh: 24-agent 2-round Qwen2.5-VL visual-symptom swarm with a shared blackboard and cross-talk (support / challenge / withdraw) and a five-step CoT consolidator.
4. Step 3 (LOCAL) — scripts/sh_03_validate_local.sh: Claude + WebSearch verifier over each unverified delta against extension / APS / CABI / peer-reviewed sources.
5. Step 4 (NOVA) — scripts/sh_04_train_encoder_nova.sh: BioCAP-style ViT-B/16 dual-projector CLIP encoder fine-tuned on Bugwood + KB-grounded captions, warm-started from BioCLIP. Output checkpoint is the pathomeood_v1 encoder consumed by step 5.
6. Step 5 (LOCAL) — scripts/sh_05_tabpfn_local.sh: seven frozen encoders (six off-shelf + your step-4 pathomeood_v1 as variant T15) emit [image_emb | caption_emb | crop_text_emb | state_text_emb] features. TabPFN classifies over the 15-variant ablation matrix; Grad-CAM (BioCAP §C.3 reproduction) emits qualitative figures and the energy-pointing-game score (with –bbox-csv).

Steps 0–3 build PathomeDB. Steps 4–5 are the PathomeOOD evaluation harness: step 4 contributes a domain-specialised encoder; step 5 evaluates it alongside six off-shelf encoders.

F.2. PathomeDB seed summary

Metric	Value
Profiles	25
With canonical text	25
With regional KB	0
Regional deltas	0

Table 15. PathomeDB seed summary.

F.3. Phase 0R routed-swarm trace stats

(Trace stats not yet generated; run bash scripts/viz_traces.sh.)

F.4. Observe **training curves**

(Observe training history not yet generated; run bash scripts/viz_observe.sh.)

F.5. Observe **held-out evaluation**

(Observe eval not yet generated; run bash scripts/viz_observe.sh.)